

Ligjerata 2a

Kuptimi i vektorit të shpejtësisë dhe nxitimit

$$\langle v \rangle = v_m = \frac{\Delta s}{\Delta t} \rightarrow \text{rruga.}$$

$$\Delta t \rightarrow \text{kooha.}$$

$\Delta t \rightarrow 0$ $dt \rightarrow$ elementi i kohës

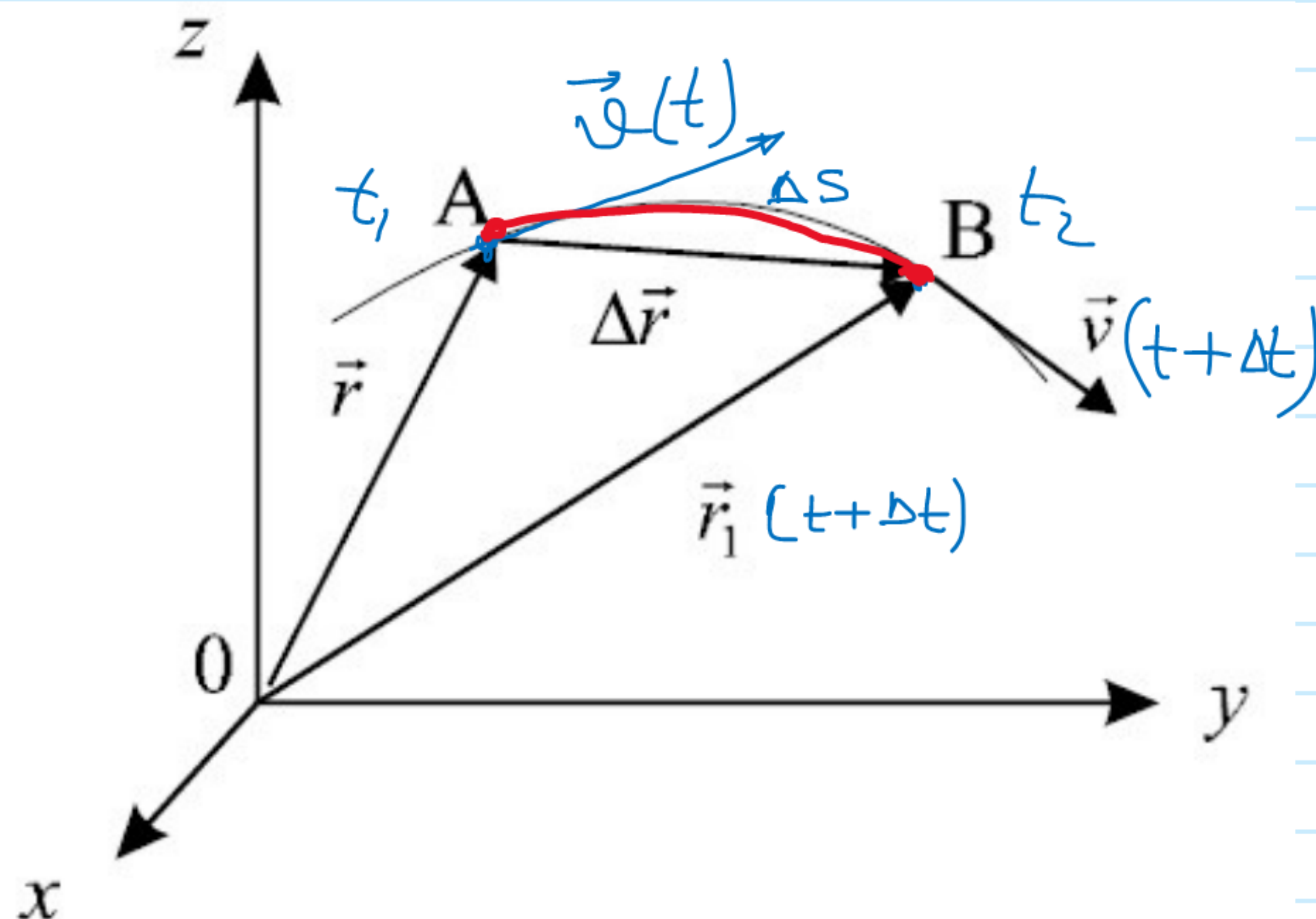
$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt} = \vec{s}$$

$\overline{AB} \rightarrow AB$

$\Delta s \rightarrow \Delta r$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$



$$\vec{v} = \frac{d}{dt} (x\vec{i} + y\vec{j} + z\vec{k}) = \frac{d}{dt} (x\vec{i}) + \frac{d}{dt} (y\vec{j}) + \frac{d}{dt} (z\vec{k})$$

$$\vec{v} = \frac{dx}{dt} \vec{i} + \frac{dy}{dt} \vec{j} + \frac{dz}{dt} \vec{k}$$

$$\frac{dx}{dt} = v_x ; \frac{dy}{dt} = v_y ; \frac{dz}{dt} = v_z$$

komponentet skalare të $\vec{v}$

$$\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$$